


Random Forest

DT → easy to use
easy to build
easy to interpret

Train dataset ✓
New data X } → Overfitting
[Low bias]

RF combines the simplicity of DT with flexible results
better accuracy.

Step 1: Creating a bootstrapped data.

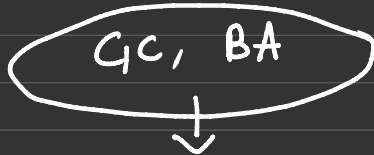
Original Dataset				
Chest Pain	Good Blood Circ.	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	Yes	167	Yes

→ Bootstrapped data.

CP	BC	BA	W	HD
Y	Y	Y	180	Y
N	N	N	125	N
Y	N	Y	167	Y
Y	N	Y	167	Y

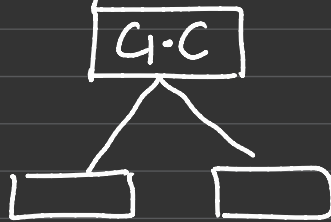
Step 2: Create a DT using bootstrapped dataset.

Instead of picking all variable to decide root node, we will randomly select columns



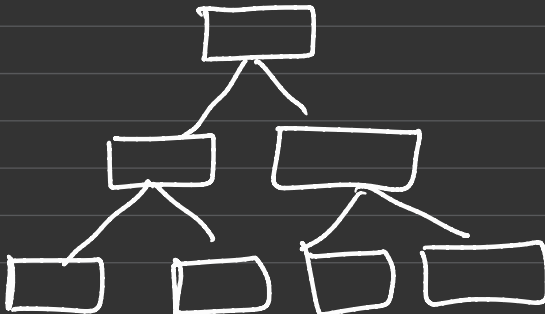
Find Gini Impurity
Compare and pick the best

let.



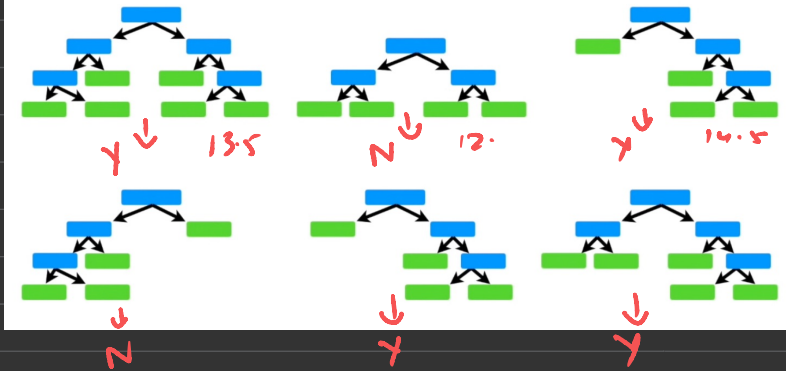
For branches also randomly I will pick 2

C-P, W



- ①

Now go back to Step 1 and repeat: Make a new bootstrapped dataset and build a tree considering a subset of variables at each step.



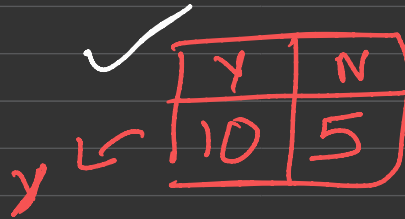
* The variety is what makes random forests more effective than individual DT.

Well, first we get a new patient...

Chest Pain	Good Blood Circ.	Blocked Arteries	Weight	Heart Disease
Yes	No	No	168	??

??
↓

1) Run this data through all the trees.



For regressor problem: Avg will be taken.

Bootstrapping the data + using the aggregation to make a decision is called **Bagging**.

② — The data which is not used in bootstrapping from original data is known as **Out of Bag dataset**

— Typically $\frac{1}{3}$ rd of your original dataset doesnot come under bootstrapping.

The proportion of out of Bag samples that were incorrectly classified is
is **Out of Bag Error**